

Predicting Cardiovascular Disease Using Machine Learning

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Abstract—Cardiovascular diseases (CVDs) remain a leading global cause of death, creating the need for early detection. This project applies logistic regression to predict the risk of Myocardial Infarction or Coronary Heart Disease (MICHHD) using Behavioral Risk Factor Surveillance System (BRFSS) data. After preprocessing the data the model was trained with gradient descent and L2 regularisation, then evaluated using ROC metrics. Results show strong predictive accuracy of 86.9% and an F1 score of 0.428, highlighting the potential of interpretable machine learning for large-scale health monitoring.

I. INTRODUCTION

Cardiovascular diseases (CVDs) are a leading cause of mortality in developed nations, accounting for millions of largely preventable deaths each year. Myocardial Infarction and Coronary Heart Disease (MICHHD) are among the most common and often fatal without early detection. Identifying high-risk individuals early can significantly improve prevention and outcomes. This project aims to develop a machine learning model to predict MICHHD risk using data from the Behavioral Risk Factor Surveillance System (BRFSS). Each record represents an individual's health profile, and the task is to predict whether that person is likely to develop MICHHD. The project applies logistic regression, focusing on model interpretability and practical use for population-level health monitoring.

II. MODEL AND METHODS

A. Dataset and preprocessing

The BRFSS dataset is a large-scale, phone-based survey collecting self-reported health data across domains such as health status, healthcare access, chronic conditions, demographics, lifestyle, and preventive behaviors. Dataset comprises a heterogeneous mix of binary, categorical, and numerical attributes including administrative and derived variables aggregating by analysts from raw responses. To render the data suitable for logistic regression:

- Administrative variables, including phone numbers and call information, were dropped.
- Categorical variables were transformed using one-hot encoding.
- Missing values were handled by adding indicator columns that mark missing entries, allowing the model to treat the lack of data itself as informative.
- Strong 10:1 class imbalance (low MICHHD prevalence) was addressed by introducing class weights. Misclassifications of the minority class were penalized more heavily, reducing bias toward the majority class.

- Categories with more than ten unique values were treated as continuous features, while those with fewer were encoded as categorical variables.

The dataset was randomly split into a training set (80%) and test set (20%).

B. Model Overview

Project employs **Logistic Regression**, a robust, classical binary classifier. The model estimates the probability that an individual develops MICHHD given an input feature vector $\mathbf{x}$:

$$P(y = 1 | \mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \quad (1)$$

where $\mathbf{w}$ denotes the learned weight vector, $\mathbf{x}$ the input feature vector, and b the bias term.

Model hyperparameters, including learning rate, regularization strength, class weighting, were optimized via grid search. The optimal decision threshold was selected using best F1 score. Model weights were learned using Gradient Descent (GD) to minimize the negative log-likelihood loss. To improve generalization, L2 regularization with coefficient $\alpha = 0.1$ was applied, penalizing large weights and preventing overfitting, resulting in the loss function defined in Eq. 2. Training used early stopping with a patience of 20 epochs, halting optimization when validation loss stopped improving.

$$\mathcal{L}(\mathbf{w}) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] + \alpha \|\mathbf{w}\|_2^2 \quad (2)$$

where $\hat{y}_i$ denotes the predicted probability for sample i .

III. RESULTS AND DISCUSSION

The dataset was large enough for the law of large numbers to apply, making an 80/20 split appropriate without additional balancing or stratification. Grid search indicated that the best accuracy was achieved with a class weight of 10, meaning the loss penalty for each class was scaled by its respective weight. This ratio directly reflects the class distribution:

$$\frac{\# \text{class}(-1)}{\# \text{class}(1)} \approx 10 \quad (3)$$

Where $\text{class}(-1)$ means not at risk of MICHHD and $\text{class}(1)$ means at risk.

The loss curve in figure 1 shows stable and consistent convergence of both training and validation loss. The model quickly reduces loss within the first few iterations and gradually

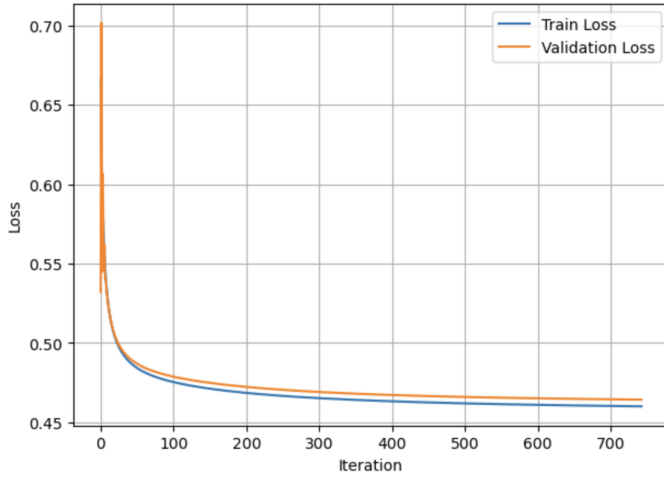


Fig. 1. Training and validation loss. Blue line indicating train and yellow indicating validation loss

approaches a plateau around 0.45, indicating effective learning without signs of overfitting. The close alignment between training and validation loss suggests good generalization and stable optimization throughout training.

Our model achieved an accuracy of 86.9%, which reflects overall performance across both classes and is not informative about minority class prediction. The F1 score provides a clearer view of stroke prediction capability, with a value of 0.428, precision of 0.351, and recall of 0.551. This indicates that roughly half of the individuals predicted as positive were actual stroke cases.

The most contributing features for MICHHD prediction as shown in figure 2 were prior stroke history (CVDSTRK3_1.0), male gender (SEX_1.0), and older age (AGE80, AGE5YR) had the strongest positive weights, indicating higher disease risk. Good general health and absence of chronic conditions, such as COPD, or kidney disease (e.g., CHCCOPD1_2.0, CHCKIDNY_2.0) had large negative weights, reducing predicted risk. This pattern aligns with known cardiovascular risk factors - age, sex, and prior stroke remain the dominant predictors, while indicators of good health are protective. Big importance can be seen in GENHLTH_1.0–5.0, ranking health from poor (5) to excellent (1). This may reflect both objective and subjective factors, as individuals with a history of MICHHD are likely to report poorer overall health and feel worse due to their condition. A surprising factor was MARITAL_5.0 (never married), as individuals who were never married showed a coefficient ranking among the top ten predictors, suggesting a potential link between social or lifestyle factors and MICHHD risk.

IV. SUMMARY

This project showed that a simple logistic regression model can effectively predict MICHHD risk using large-scale BRFSS survey data. Despite limited computation and restricted feature tuning, the model achieved 86.9% accuracy and provided interpretable insights into key risk factors. Age, sex, prior

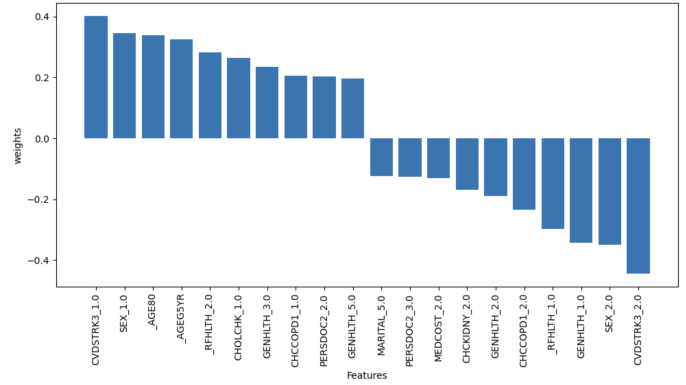


Fig. 2. Twenty highest weight coefficients of logistic regression. Variables and encoded values: CVDSTRK3_1.0/2.0 (ever told you had a stroke; 1 = yes, 2 = no), SEX_1.0/2.0 (biological sex; 1 = male, 2 = female), AGE80 (respondent age capped at 80), AGE5YR (age grouped in 5-year intervals), RFHLTH_1.0/2.0 (general health; 1 = good or better, 2 = fair or poor), CHOLCHK_1.0 (Cholesterol checked in the last 12 months), GENHLTH_1.0–5.0 (self-rated health; 1 = excellent, 2 = very good, 3 = good, 4 = fair, 5 = poor), CHCCOPD1_1.0/2.0 (told you have COPD, emphysema, chronic bronchitis; 1 = yes, 2 = no), PERSDOC2_2.0/3.0 (number of personal health provider; 2 = more than 1, 3 = 0), MARITAL_5.0 (marital status; 5 = never married), MEDCOST_2.0 (could not afford to see a doctor in the last 12 months; 2 = no), CHCKIDNY_2.0 (kidney disease; 2 = no). Positive coefficients increase predicted disease risk, negative coefficients reduce it.

stroke, and self-rated health had the strongest impact, consistent with known medical evidence. The influence of subjective health perception and marital status among top predictors suggests that both physiological and social factors play a role in cardiovascular risk. Overall, the project demonstrates that interpretable linear models can extract meaningful patterns even from complex population health data.

V. LIMITATIONS AND UNSUCCESSFUL APPROACHES

A major limitation of the project was the slow computation time of the model. Since the model was implemented from scratch, it lacked the optimizations of dedicated libraries, resulting in long training times that were only partially mitigated by simplifying some procedures. The dataset split was fixed without shuffling, with 80% used for training and 20% for testing, so the same subsets were always applied. Other Models, like Support Vector Machines, were implemented but discarded due to worse performance overall. Different feature selection approaches were tested using various subsets of variables. These included using all variables, only derived variables, and manually filtered subsets where highly correlated features were removed. However, the best results were obtained when using all variables, both calculated and raw. Data cleaning remained limited, and correlations between variables likely reduced model efficiency.